

COURSE DESCRIPTION FORM

INSTITUTION National University of Computer and Emerging Sciences

PROGRAM (S) TO BE Computer Science

EVALUATED _____

A. Course Description

(Fill out the following table for each course in your computer science curriculum. A filled-out form should not be more than 2-3 pages.)

Course Code	CS3002	
Course Title	Information Security	
Credit Hours	3	
Prerequisites by Course(s) and Topics	Computer Networks (CS3001), Operating Systems (CS2006)	
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	Assessment with the weight.	
	Assessment Type	Weight
	Assignments	10
	Quiz	10
	Mid-Term	25-30
	Project	10
	Final	40-45
Course Coordinator	Dr. Syed Mohammad Irteza	
URL (if any)	IS (Fall 2024) BDS-7A (google.com) IS (Fall 2024) BDS-7B (google.com) IS (Fall 2024) BSE-7A (google.com)	
Current Catalog Description	Introduction to Information security, The CIA Triad: Confidentiality Integrity and Availability, Information security Models, Security compliance laws and regulations, Governance frameworks, Risk analysis, Security architectures, Malware classification. types of malware. Cryptography, Database & web security, Network security, Security policies	
Textbook (or Laboratory Manual for Laboratory	Cryptography and Network Security: Principles and Practice, William Stallings Principles of Information Security, Michael E. Whitman, Herbert J. Mattord Computer Security: Principles and Practice, William Stallings, Lawrie Brown	

Courses)	Hands-on Labs for Security Education, by SEED labs	
Reference Material	Computer Security Fundamentals (second edition): Chuck Easttom	
Topics Covered in the Course, with Number of Lectures on Each Topic (assume 15-week instruction and one-hour lectures)	Timeline	Content Covered
	Lecture 1	Course Introduction - Introducing syllabus, policies, and projects. - An overview of basic information security principles (with practical examples): confidentiality, integrity, availability, authentication, authorization and non-repudiation. - Components of an Information System
	Lecture 2	Security Design Principles Discussion and evaluation of following primitives: Least-privilege, fail-safe defaults, complete mediation, separation of privilege, economy of mechanism, open design
	Lecture 3	Cryptography Introduction to Cryptography: Symmetric cipher model, Substitution techniques (Caesar cipher, Monoalphabetic cipher)
	Lecture 4	Cryptography-II Substitution techniques (Vigenère cipher, One-time pad) Transposition techniques (Rail fence cipher, Row transposition cipher)
	Lecture 5	Cryptography-III Block cipher structure and design principle, Feistel cipher structure, the data encryption standard, DES (encryption, key generation)
	Lecture 6	Cryptography-IIV AES structure, transformation, key expansion mechanism, AES example and implementation Stream ciphers introduction
	Lecture 7	Cryptography-V Introduction to Public Key cryptography RSA: principles, RSA algorithm Diffie-Hellman key exchange algorithm with example, Man-in-the-middle attack in Diffie-Hellman
	Lecture 8	Cryptography-VI Hash functions, applications, Hash properties (preimage resistant, second preimage resistant, collision resistant) Message authentication code, requirements & properties of

	MAC HMAC algorithm & structure
Lecture 9	Cryptography-VII Digital Signature, requirements & properties of DS Public key infrastructure (PKI), elements of PKI, X.509, Digital certificates
Lecture 10	Revision
First Mid-term Exam	
Lecture 11	Software Security Malware, types of malware (virus, worms, trojan horse, adware, spyware, backdoor, ransomware, rootkits, bootkits), malware analysis & countermeasures
Lecture 12	Software Security-II Control Hijacking: Integer overflow String format vulnerabilities & countermeasures
Lecture 13	Software Security-III Control Hijacking: Buffer overflow Countermeasures
Lecture 14	Database Security Basics SQL Injection Attack, techniques, types of attack Countermeasures, database access control
Lecture 15	Database Security-II Database inference attacks & counter measures Database encryption methods
Lecture 16	Web Security Background Cross Site Request Forgery (CSRF) Attack Countermeasures (STP, origin header, referrer header)
Lecture 17	Web Security-II Cross Site Scripting (XSS) Attack Types of XSS (reflected, stored, DOM based) countermeasures (encoding, validation, input handling contexts, secure input handing)

	Lecture 18	User Authentication Types (password, biometric, symmetric/asymmetric) Kerberos (overview, key exchange protocol)
	Lecture 19	Access Control Access control policies Discretionary and Role-based Access Control
	Lecture 20	Revision
	Second Mid-term Exam	
	Lecture 21	Network Security Secure Socket Layer (SSL) SSL certificate, architecture, handshake
	Lecture 22	Network Security-II IP security (IPSec) IPsec modes (transport, tunnel), architecture, AH, ESP
	Lecture 23	Network Security-III Intrusion Detection Systems (IDS) Components of IDS, classification of IDS (anomaly, signature, hybrid), types (host-based, network based)
	Lecture 24	Network Security-IV Firewalls Types of firewall (packet-filtering, stateful packet inspection, application proxy, circuit-level proxy) Location of firewall
	Lecture 25	Theoretical models of Access Control Confidentiality policies (BLP model) Integrity policies (Biba Model) Integrity policies (Clark-Wilson model) Hybrid policies (Chinese Wall model)
	Lecture 26	Cybercrime Laws and Ethics Pakistan cybercrime act and the role of investigative agencies. Ethical perspective of research studies and experimentation (data privacy and anonymization techniques). Intellectual property, copyright, patent, trade secret.

	Lecture 27 - onwards	Revision & Project Evaluations		
	Final Examination			
Laboratory Projects/Experiments Done in the Course				
Programming Assignments	A programming assignment where students are expected to develop an application with a focus on identifying vulnerabilities and implementing mechanisms to address them.			
Class Time Spent on (in credit hours)	Theory	Problem Analysis	Solution Design	Social and Ethical Issues
	40	25	25	10
Oral and Written Communications	Every student is required to submit at least __2__ written reports for the given assignments and to make __1__ oral presentations of typically __10__ minute's duration for the project. Include only material that is graded for grammar, spelling, style, and so forth, as well as for technical content, completeness, and accuracy.			

Instructor Name: Dr. Syed Mohammad Irteza



Instructor Signature: